MIRZO ULUG‘BEK NOMIDAGI O‘ZBEKISTON MILLIY UNIVERSITETI
Amaliy matematika va kompyuter tahlili kafedrasi

MATLATIPOV SANATBEK

O‘zbek tili gaplarini neyron tarmoqlar asosida sintaktik tahlil qilishning matematik modeli

05.01.07 – Matematik modellashtirish. Sonli usullar va dasturlar majmui

Kafedra ilmiy seminari

Ilmiy rahbar: f.-m.f.d., professor Aripov Mersaid Mirsidikovich

Toshkent – 2026

Mavzuning dolzarbligi

O‘zbek tili — 35 mln dan ortiq kishi so‘zlashadigan **agglyutinativ** turkiy til.

Gapning **sintaktik tuzilishi** mashina tarjimasi, axborot qidiruvi va savol-javob tizimlari uchun asos.

Kam resursli muhit: Universal Dependencies (UD) tizimida o‘zbek tili uchun bor-yo‘g‘i **1 184 ta gap** sintaktik annotatsiyalangan.

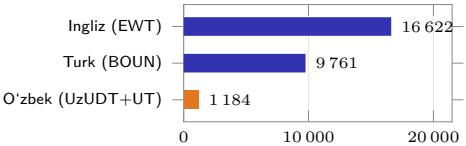
Kelishik, egalik, son, zamon va shaxs ma’nolari **qo‘shimchalar zanjiri** bilan ifodalanadi ⇒ so‘z shakllari ko‘p, lug‘at siyrak.

Kichik tanlanmada ishonchli ishlaydigan, **matematik asoslangan** model va **sonli usullar** (aniq dekodlash, turg‘un o‘qitish) zarur.

So‘z shakli: morfemalar va BPE qism so‘zlari

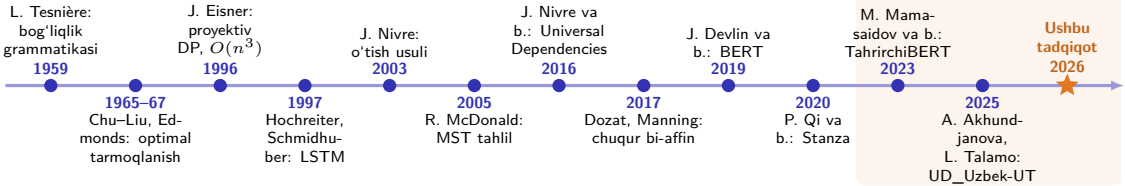


UD korpuslari hajmi (gaplar soni)



Manba: universaldependencies.org. BPE bo‘linishi — TahrirchiBERT tokenizatorining haqiqiy natijasi.

Muammoning o‘rganilganlik darajasi



O‘zbekiston olimlari

M. M. Aripov, M. X. Xakimov, G‘. R. Matlatipov, N. Abdurakhmonova, M. S. Sharipov, E. R. Quriyozov, B. B. Elov, U. I. Salaev, M. Mamasaidov

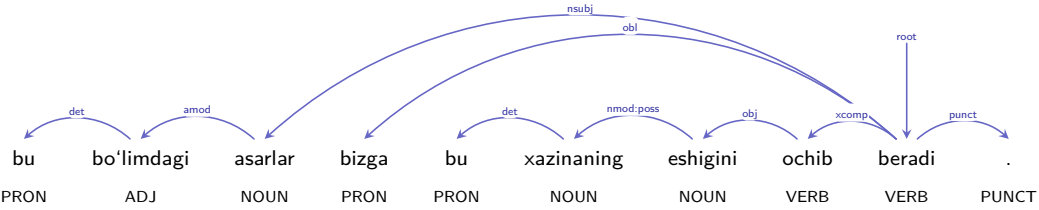
Xorijiy va MDH olimlari

L. Tesnière, J. Edmonds, R. Tarjan, J. Eisner, J. Nivre, R. McDonald, T. Dozat, C. D. Manning, P. Qi, J. Devlin, M.-C. de Marneffe, D. Zeman, M. Straka, G. Eryiğit, U. A. Tukeyev (Qozog‘iston)

Hal etilmagan masala

O‘zbek tili uchun (i) UD sintaktik korpuslari juda kichik va heterogen; (ii) BPE qism so‘zlaridan **agglyutinativ so‘z vektorini** hosil qilish operatori asoslanmagan; (iii) aniq (MST) va ochko‘z dekodlash hamda modelning **turg‘unligi** tizimli o‘rganilmagan.

Masalaning qo‘yilishi



Ta’rif 1 (bog‘liqlik daraxti)

$x = (w_1, \ldots, w_n)$ — gap, $w_0 = \text{ROOT}$;
$y = \{(h_i, i, r_i)\}_{i=1}^{n}$: $h_i \in \{0, \ldots, n\} \setminus \{i\}$ — **hokim** so‘z, $r_i \in \mathcal{R}$ ($|\mathcal{R}| = 39$).

$y \in \mathcal{T}(x) \iff G_y = \big(\{0, \ldots, n\}, \{h_i \to i\}\big)$ — 0 ildizli **yoyiluvchi arborestsensiya**, $|\{i : h_i = 0\}| = 1$.

Lemma 1 (A. Cayley formulasi asosida)

$|\mathcal{T}(x)| = n^{n-1}$: $n = 10 \Rightarrow 10^9$; $\bar{n} = 11{,}4 \Rightarrow \approx 10^{11}$.

Isbot g‘oyasi: ildizli belgilangan daraxtlar bilan biyeksiya; Cayley: $n \cdot n^{n-2}$.

Masala (P)

Dekodlash: $\hat{y} = \arg\max_{y \in \mathcal{T}(x)} \log P_\theta(y \mid x)$

O‘qitish: $\hat{\theta} = \arg\min_\theta \widehat{R}(\theta)$,
$\widehat{R}(\theta) = -\frac{1}{|\mathcal{D}|} \sum_{(x,y) \in \mathcal{D}} \log P_\theta(y \mid x)$

Teglash: $\hat{t}_i = \arg\max_t P_\varphi(t \mid x, i), \quad t \in \mathcal{U} \times \mathcal{X} \times \mathcal{F}$

Sifat: $\text{UAS} = \frac{1}{N} \sum_i \mathbf{1}[\hat{h}_i = h_i]$,
$\text{LAS} = \frac{1}{N} \sum_i \mathbf{1}[\hat{h}_i = h_i \wedge \hat{r}_i = r_i]$

Tadqiqotning maqsadi, vazifalari, obyekti va predmeti

Tadqiqotning maqsadi

Kam resursli sharoitda o‘zbek tili gaplarini avtomatik morfosintaktik tahlil qilishning neyron tarmoqli **matematik modelini**, uni o‘qitish va dekodlashning **sonli usullarini** ishlab chiqish hamda ularni **dasturiy majmua** sifatida amalga oshirish.

Tadqiqotning vazifalari

1. UD standartidagi oltin standart sintaktik korpus (**UzUDT**) yaratish va uni statistik tahlil qilish;
2. tahlil masalasini arborestsensiyalar to‘plamida **optimallashtirish masalasi** sifatida formallashtirish;
3. agglyutinativ morfologiyani hisobga oluvchi **qism so‘z vektorlarini birlashtirish operatorini** ishlab chiqish;
4. bi-affin baholash, MST dekodlash va korpuslararo o‘qitishning **sonli usullarini** tadqiq qilish;
5. aniqlik va **turg‘unlikni** tajribada baholash; dasturiy majmuani yaratish.

Obyekt va predmet

Obyekt: o‘zbek tilidagi matnlar va ularning UD sintaktik korpuslari (UzUDT, UT).

Predmet: sintaktik tahlilning graf-ehtimollik modeli, uni o‘qitish va dekodlashning sonli usullari, dasturiy majmuasi.

Usullar

Graflar nazariyasi, kombinatorik va stoxastik optimallashtirish, ehtimollar nazariyasi, matematik statistika, neyron tarmoqlar.

Tadqiqotning ilmiy yangiligi

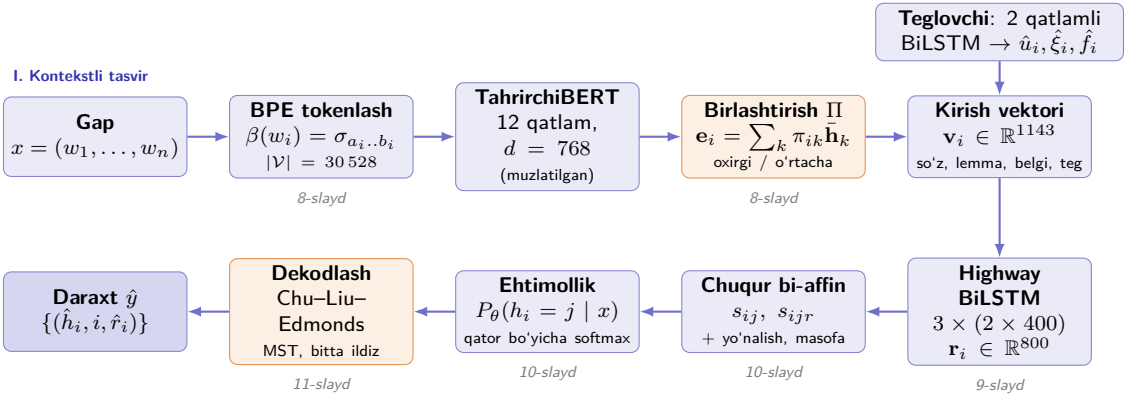
1. O‘zbek tili uchun UD standartidagi oltin standart sintaktik korpus — **UzUDT** (684 gap, 7 582 token, annotatorlararo kelishuv $\kappa = 0{,}91$–$0{,}95$) yaratilgan, UD v2.17 tarkibida ochiq e’lon qilingan; uning statistik xossalari (noproyektivlik, yoy uzunligi, hokim yo‘nalishi) aniqlangan. [13-sl.]

2. Bog‘liqlik tahlili masalasi yoyiluvchi arborestsensiyalar to‘plamida ehtimollikni maksimallashtirish masalasi sifatida formallashtirilgan; agglyutinativ morfologiyani hisobga oluvchi **«oxirgi qism so‘z»** birlashtirish operatori taklif etilgan va u o‘rtacha qiymat operatoriga nisbatan LAS ni **statistik ahamiyatli** oshirishi ko‘rsatilgan ($+2{,}6$ punkt, $p \approx 0{,}002$). [8, 10, 14-sl.]

3. Korpuslararo empirik risk bo‘yicha stoxastik optimallashtirish (Adam $\rightarrow$ AMSGrad, erta to‘xtatish) va **Chu–Liu–Edmonds** algoritmi bilan aniq dekodlashning sonli sxemasi ishlab chiqilgan; uning ochko‘z o‘tish usuliga nisbatan ustunligi (bir xil test to‘plamida LAS $+8{,}7$) va orfografik g‘alayonlarga **turg‘unligi** miqdoriy baholangan. [11, 12, 15-sl.]

4. Stanza kutubxonasining teglovchi va parseriga kontekstli vektorlarni ulaydigan modullar (`--bert_model`, `--bert_pooling`), spaCy uchun o‘zbek tili moduli, baholash va bootstrap vositalaridan iborat **dasturiy majmua** yaratilgan; o‘qitilgan modellar ochiq e’lon qilingan. [16-sl.]

Matematik modelning umumiy sxemasi



Ikki bosqichli quvur: avval teglovchi o'qitiladi, so'ng parser kirishiga UPOS/XPOS/UFeats teglari beriladi (A tajribalarda — bashorat qilingan, B da — oltin); FastText variantida $\mathbf{e}_i$ o'rniga $\mathbf{W}_p \mathbf{p}_i \in \mathbb{R}^{125}$.

Qism so‘z vektorlarini so‘z darajasiga o‘tkazish operatori

Ta’rif 2 (BPE va qatlamlar)

$\beta:\ w_i \mapsto (\sigma_{a_i}, \ldots, \sigma_{b_i}),\ \ m_i = b_i - a_i + 1$ — qism so‘zlar soni.
Kontekstli tasvir (Stanza sxemasi):
$\bar{\mathbf{h}}_k = \frac{1}{4}\sum_{\ell=9}^{11} \mathbf{H}_k^{(\ell)} \in \mathbb{R}^{768}$.

Ta’rif 3 (birlashtirish operatori)

$\mathbf{E} = \Pi\,\bar{\mathbf{H}},\ \ \mathbf{e}_i = \sum_{k=1}^{M} \pi_{ik}\bar{\mathbf{h}}_k,\ \ \Pi \in \mathbb{R}^{n\times M}$ — **qator-stoxastik** matritsa:

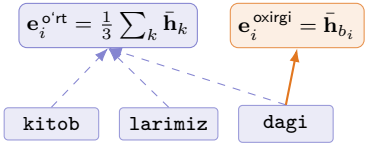
$$\pi_{ik}^{\text{oxirgi}} = \mathbf{1}[k = b_i], \qquad \pi_{ik}^{\text{o‘rt}} = \frac{\mathbf{1}[a_i \le k \le b_i]}{m_i}.$$

Gipoteza (suffiks lokalizatsiyasi)

O‘zbek tilida grammatik ko‘rsatkichlar so‘zning **o‘ng chetida** joylashadi $\Rightarrow \bar{\mathbf{h}}_{b_i}$ kelishik, shaxs-son ma’lumotini to‘playdi; o‘rtacha qiymat operatori bu signalning ulushini $1/m_i$ gacha kamaytiradi.

TahrirchiBERT tokenizatorining haqiqiy natijasi

kitoblarimizdagi	kitob	larimiz	dagi	
o‘quvchilarimizdagi	o	‘	quvchi	larimiz	dagi
maktablarimizning	maktab	larimizning		
ko‘rganmisiz	ko	‘	rgan	misiz	
xazinaning	xazina	ning		



o‘/g‘ dagi tutuq belgisi (‘) alohida qism so‘zga ajraladi.

Kontekstni hisobga oluvchi kodlovchi: highway BiLSTM

Kirish vektori

$$\mathbf{v}_i = \left[\mathbf{e}_i^{w};\ \mathbf{e}_i^{\text{lem}};\ \mathbf{e}_i^{\text{upos}} + \mathbf{e}_i^{\text{xpos}};\ \sum_k \mathbf{e}_i^{f_k};\ \mathbf{W}_c\mathbf{c}_i;\ \mathbf{e}_i\right] \in \mathbb{R}^{1143}$$

so'z 75 + lemma 75 + teglar 50 + belgilar 50 + harf-LSTM 125 + BERT 768

LSTM yacheykasi (S. Hochreiter, J. Schmidhuber, 1997)

$$\begin{pmatrix}\mathbf{i}_t\\ \mathbf{f}_t\\ \mathbf{o}_t\\ \mathbf{g}_t\end{pmatrix} = \begin{pmatrix}\sigma\\ \sigma\\ \sigma\\ \tanh\end{pmatrix}\left(\mathbf{W}\begin{bmatrix}\mathbf{v}_t\\ \mathbf{h}_{t-1}\end{bmatrix} + \mathbf{b}\right),$$

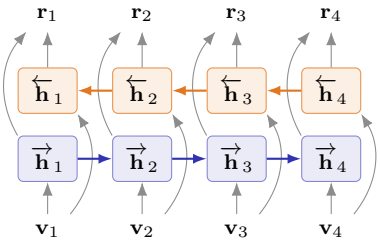
$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \mathbf{g}_t,$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t).$$

Ikki yo'nalish va highway bog'lanish, $\ell = 1, 2, 3$

$$\mathbf{z}_t^{(\ell)} = \left[\overrightarrow{\mathbf{h}}_t^{(\ell)}; \overleftarrow{\mathbf{h}}_t^{(\ell)}\right] + \sigma\left(\mathbf{G}^{(\ell)}\mathbf{z}_t^{(\ell-1)}\right) \odot \tanh\left(\mathbf{W}_H^{(\ell)}\mathbf{z}_t^{(\ell-1)}\right),$$

$$\mathbf{r}_i = \mathbf{z}_i^{(3)} \in \mathbb{R}^{800}$$



Gradient oqimi

To'g'ridan-to'g'ri yo'l bo'yicha $\dfrac{\partial \mathbf{c}_t}{\partial \mathbf{c}_{t-1}} = \operatorname{diag}(\mathbf{f}_t)$: unutish darvozasi uzoq bog'lanishlarda gradientning so'nishini yumshatadi.

Teglovchi: 2 qatlam, 2×200; parser: 3 qatlam, 2×400.

Chuqur bi-affin baholash va ehtimollik modeli

Ta'rif 4 (chuqur bi-affin baholovchi)

$\mathbf{a}_i^d = \mathrm{ReLU}(\mathbf{W}_d \mathbf{r}_i + \mathbf{b}_d)$, $\mathbf{a}_j^h = \mathrm{ReLU}(\mathbf{W}_h \mathbf{r}_j + \mathbf{b}_h) \in \mathbb{R}^{400}$,

$s_{ij} = [\mathbf{a}_i^d; 1]^\top \mathbf{U} [\mathbf{a}_j^h; 1]$, $\mathbf{U} \in \mathbb{R}^{401 \times 401}$; munosabat: s_{ijr}, $\mathbf{U}^{(r)}$.

Lemma 2 (bi-affin forma — bilinear forma)

$\mathbf{U} = \begin{bmatrix} \mathbf{U}_{11} & \mathbf{u}_{12} \\ \mathbf{u}_{21}^\top & u_{22} \end{bmatrix}$ bo'lsa, $s_{ij} = \mathbf{a}_i^{d\top} \mathbf{U}_{11} \mathbf{a}_j^h + \mathbf{u}_{21}^\top \mathbf{a}_j^h + c_i$,

$c_i = \mathbf{a}_i^{d\top} \mathbf{u}_{12} + u_{22}$.

softmax siljishga invariant $\Rightarrow$ c_i ta'sir qilmaydi; $\mathbf{u}_{21}^\top \mathbf{a}_j^h$ — «hokimlik apriori».

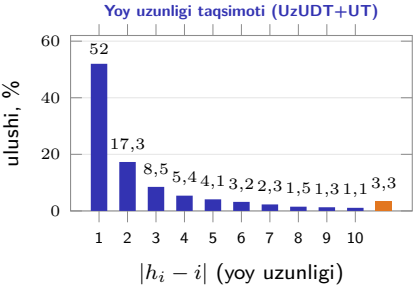
Ehtimollik modeli (lokal normallash)

$\tilde{s}_{ij} = s_{ij} + \log \sigma\big(\lambda_{ij} \operatorname{sgn}(j-i)\big) - \log\left(1 + \frac{1}{2}(|j-i| - \delta_{ij})^2\right)$

2-had — yoy yo'nalishi, 3-had — masofa uchun **Koshi yadrosi** ($\delta_{ij} \geq 1$)

$P_\theta(h_i = j \mid x) = e^{\tilde{s}_{ij}} \Big/ \sum_{k \neq i} e^{\tilde{s}_{ik}}$,

$P_\theta(r_i = r \mid x, h_i = j) = \mathrm{softmax}_r\,(s_{ijr})$.



84,6 %
hokim o'ngda (SOV)

$\bar{d} = 2{,}72$
o'rtacha yoy uzunligi

Tinish belgilari va ROOT yoylarisiz, 13 512 token. Og'ir dumli taqsimot masofa uchun Koshi tipidagi yadro tanlovini asoslaydi.

Sonli usul (1): maksimal yoyiluvchi arborestsensiya — Chu–Liu–Edmonds

Teorema 1 (Chu, Liu, 1965; Edmonds, 1967)

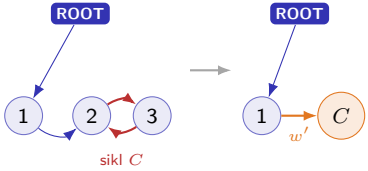
$G = (V, A, w)$ — to‘liq yo‘naltirilgan graf, $w_{ji} = \log P_\theta(h_i = j \mid x)$. Model yoylar bo‘yicha faktorlanganda

$$\hat{y} = \arg\max_{y \in \mathcal{T}(x)} \sum_{(j \to i) \in y} w_{ji}$$

maksimal og‘irlikdagi yoyiluvchi arborestsensiyadir; CLE algoritmi uni **aniq** topadi: $O(n^3)$, R. Tarjan (1977) amalga oshirishida $O(n^2)$.

Algoritm: ochko‘z tanlov + sikl kontraksiyasi

1) har bir i uchun $h_i = \arg\max_j w_{ji}$; **2)** sikl bo‘lmasa — yechim optimal; **3)** sikl C bitta uchga qisqartiriladi: $w'_{j \to C} = \max_{i \in C} \left(w_{ji} - w_{h_i i} \right)$, so‘ng rekursiya va qayta yoyish.
Bitta ildiz sharti: nomzod ildizlar bo‘yicha CLE qayta yechiladi.



(a) ochko‘z tanlov **(b) kontraksiya**

n	$\lvert\mathcal{T}(x)\rvert = n^{n-1}$	CLE: $\sim n^2$
5	625	25
10	10^9	100
20	$5{,}2 \cdot 10^{24}$	400

9,2 % gaplar noproyektiv
(109 / 1 184). Proyektiv usullar (J. Eisner, $O(n^3)$) ularni ifodalay olmaydi; CLE ifodalaydi.

Sonli usul (2): korpuslararo empirik riskni stoxastik minimallashtirish

Yo‘qotish funksiyasi (bitta gap uchun)

$$\mathcal{L}(\theta) = -\sum_{i=1}^{n} \log P_\theta(h_i^* \mid x) - \sum_{i=1}^{n} \log P_\theta(r_i^* \mid x, h_i^*) + \mathcal{L}_{\text{yo‘n}} + \mathcal{L}_{\text{mas}}$$

$\mathcal{L}_{\text{yo‘n}}$, $\mathcal{L}_{\text{mas}}$ — yoy yo‘nalishi va masofasi bo‘yicha yordamchi yo‘qotishlar

Korpuslararo empirik risk (qavariq aralashma)

$$\widehat{R}(\theta) = \lambda\, \widehat{R}_{\text{UzUDT}}(\theta) + (1-\lambda)\, \widehat{R}_{\text{UT}}(\theta), \quad \lambda = \tfrac{451}{781} \approx 0{,}58;$$

ya’ni $\widehat{P} = \lambda \widehat{P}_1 + (1-\lambda)\widehat{P}_2$

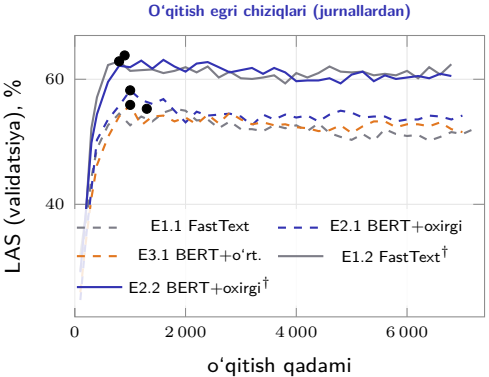
Adam → AMSGrad

$$\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1-\beta_1)\mathbf{g}_t, \quad \mathbf{v}_t = \beta_2 \mathbf{v}_{t-1} + (1-\beta_2)\mathbf{g}_t^2,$$

$$\theta_t = \theta_{t-1} - \alpha\, \hat{\mathbf{m}}_t \Big/ \left(\sqrt{\hat{\mathbf{v}}_t} + \varepsilon\right);$$

$\alpha = 3 \cdot 10^{-3}$, $(\beta_1, \beta_2) = (0{,}9;\ 0{,}95)$, $\varepsilon = 10^{-6}$, $\|\mathbf{g}_t\|_2 \le 1$; dropout 0,5; partiya 5 000 token.

To‘xtatish: 3 000 qadam yaxshilanish bo‘lmasa ⇒ AMSGrad ($\hat{\mathbf{v}}_t = \max(\hat{\mathbf{v}}_{t-1}, \mathbf{v}_t)$); yana 3 000 qadam ⇒ to‘xtash.



• — eng yaxshi nazorat nuqtasi (800–1 300 qadam); keyin haddan tashqari moslashish.

Uzuq chiziq — UzUDT, uzluksiz — UzUDT+UT; † oltin morfologik teglar bilan.

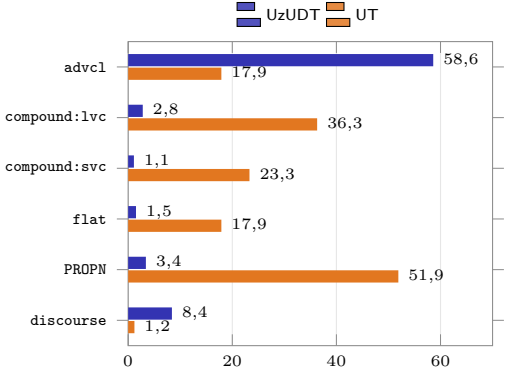
Ma’lumotlar va tajriba dizayni

	UzUDT (muallif)	**UT**
Janr	badiiy, ta’limiy	yangilik, badiiy
Gap / token	684 / 7 582	500 / 5 930
Annotatsiya	qo‘lda, 6 annotator	yarim avtomatik
Kelishuv (κ, %)	lemma 95,2; UPOS 93,7	—
UD validatsiyasi	✔ o‘tgan	✘ 92 ogohl.
Noproyektiv gaplar	11,0 %	6,8 %
Munosabat turlari	38	33 (jami 39)

Tajriba rejasi: 3×2 faktorli dizayn

Vektor / birlashtirish	UzUDT 435 / 48 / 198 gap	UzUDT+UT† 781 / 78 / 325 gap
FastText, 300	E1.1	E1.2
BERT + oxirgi	E2.1	E2.2
BERT + o‘rtacha	E3.1	E3.2‡

† bu sharoitda parserga oltin UPOS/XPOS/UFeats berilgan. ‡ E3.2 jurnali repozitoriyda yo‘q.

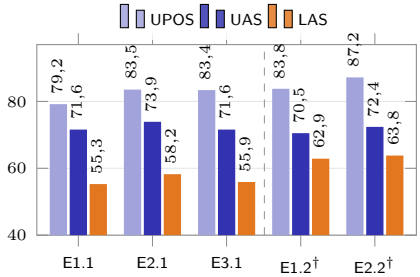


1 000 tokenga to‘g‘ri keladigan soni

Korpuslar bir-birini to‘ldiradi. Manba: data/udbase/*.conllu.

Natijalar: validatsiya to‘plami, eng yaxshi nazorat nuqtasi

		Teglash		Sintaktik tahlil	
Tajriba	**Vektorlar**	UPOS	UFeats	UAS	LAS
A. UzUDT; parserga bashorat qilingan teglar (48 gap, 644 token)					
E1.1	FastText	79,19	66,61	71,58	55,28
E2.1	BERT + oxirgi	**83,54**	**68,17**	**73,91**	**58,23**
E3.1	BERT + o‘rtacha	83,39	65,84	71,58	55,90
B. UzUDT+UT; parserga oltin teglar† *(78 gap, 851 token)*					
E1.2	FastText	83,78	70,98	70,51	62,87
E2.2	BERT + oxirgi	**87,19**	**73,21**	**72,39**	**63,81**



Xulosalar

1. Sintaktik tahlilda «oxirgi qism so‘z» operatori o‘rtacha qiymatdan **ishonchli** yaxshi.

2. TahrirchiBERT teglashni yaxshilaydi: UPOS $+3{,}4 \ldots +4{,}4$ punkt.

3. Oltin morfologiya berilganda BERT ning sintaktik ustunligi ahamiyatsiz.

Juftlashgan bootstrap, $B = 10^4$

$\Delta_b^* = \mathrm{LAS}_A(S_b^*) - \mathrm{LAS}_B(S_b^*)$; 95 % IO $= \left[\Delta^*_{(0{,}025)}; \Delta^*_{(0{,}975)}\right]$

oxirgi – o‘rt.: $+2{,}64$ $[+0{,}97; +4{,}12]$, $p \approx 0{,}002$ ✔

BERT – FT (A): $+2{,}95$ $[0{,}00; +5{,}63]$, $p \approx 0{,}03$

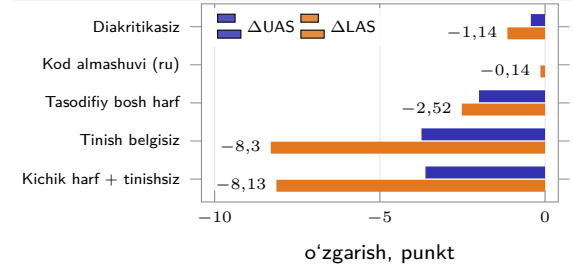
BERT – FT (B): $-1{,}29$ $[-4{,}66; +2{,}23]$ — ahamiyatsiz

† A va B bevosita taqqoslanmaydi: kiruvchi teglar (bashorat / oltin) va validatsiya to‘plamlari farqli. Bootstrap — oxirgi qadam bashoratlari bo‘yicha.

Test to‘plamida baholash: turg‘unlik va dekodlash usullarini taqqoslash

Ta’rif 5 (sezgirlik)

π — matnni buzuvchi (g‘alayon) operatori, ρ_π — o‘zgargan tokenlar ulushi; sezgirlik: $S_\pi = \mathrm{LAS}(\mathcal{D}) - \mathrm{LAS}\big(\pi(\mathcal{D})\big)$.



E2.1 modeli, UzUDT test to‘plami (198 gap, 2 101 token); toza matnda UAS 71,06, LAS 54,07. ρ_π: diakritika 8,7 %, kod almashuvi 0,2 %, bosh harf 53,3 %, tinish belgisi 19,6 %.

Aniq va ochko‘z dekodlash (bir xil test to‘plami)

	Graf + CLE Stanza, E2.1	**O‘tish (arc-eager)** spaCy, S1.1	Δ
UAS	**71,06**	67,72	$+3{,}34$
LAS	**54,07**	45,35	$+8{,}72$

Ikkalasi ham TahrirchiBERT; spaCy baholovchisi tinish belgilarini hisobga olmaydi.

Talqin

1. Global (MST) optimallashtirish ochko‘z ketma-ket qarorlardan **8,7 punkt** yaxshi: xatolar zanjir bo‘ylab tarqalmaydi.
2. Diakritika yo‘qolishiga turg‘un ($-1{,}1$), kod almashuviga deyarli befarq ($-0{,}1$).
3. Tinish belgilari — muhim tuzilmaviy signal: ularsiz LAS $8{,}3$ punktga tushadi.

Dasturiy majmua

Ma'lumotlar

UzUDT (UD v2.17, CC BY-SA 4.0) va UT

CoNLL-U o'quvchi va birlashtiruvchi

spaCy DocBin konvertori

korpus statistikasi

O'qitish

Stanza teglovchi va parseri: `--bert_model`, `--bert_pooling {last, mean}`

spaCy uchun uz til moduli

CSV / W&B jurnallari

Baholash

CoNLL 2018 metrikalari (UAS, LAS, MLAS, BLEX)

juftlashgan bootstrap

g'alayon (turg'unlik) testlari

E'lon qilish

HuggingFace: `Sanatbek/uzudt`

GitHub: `robust-parsing-uzbek`

UD rasmiy repozitoriysi

Ishga tushirish namunasi

```
python -m stanza.models.parser --mode train \
  --bert_model tahrirchi/tahrirchi-bert-base \
  --bert_pooling last \
  --train_file uz.train.in.conllu \
  --eval_file uz.dev.in.conllu
```

Texnik ko'rsatkichlar

Python, PyTorch; Stanza (teglovchi va parser modullari o'zgartirilgan), spaCy 3.8; NVIDIA RTX A6000. Eng yaxshi model 600–1 300 qadamda topiladi.

Xulosa

1. O‘zbek tili uchun oltin standart UD korpusi **UzUDT** (684 gap, 7 582 token) yaratildi va e’lon qilindi; korpusning 11 % gaplari noproyektiv, yoylarning 84,6 % ida hokim o‘ngda joylashadi.
2. Sintaktik tahlil masalasi **yoyiluvchi arborestsensiyalar** to‘plamida ehtimollikni maksimallashtirish masalasi sifatida qo‘yildi; yechimlar soni n^{n-1}.
3. BPE qism so‘zlarini birlashtirishning **qator-stoxastik operatori** kiritildi; «oxirgi qism so‘z» operatori LAS ni $+2{,}6$ punktga oshiradi ($p \approx 0{,}002$).
4. Aniq **Chu–Liu–Edmonds** dekodlash ochko‘z o‘tish usulidan LAS bo‘yicha **8,7 punkt** yuqori.
5. Model diakritika va kod almashuviga turg‘un, tinish belgilariga sezgir ($-8{,}3$ LAS).
6. **Dasturiy majmua** (Stanza–BERT, spaCy-uz, baholash vositalari) yaratildi, modellar ochiq e’lon qilindi.

Kelgusi ishlar

Barcha tajribalarni yagona bo‘linma va bashorat qilingan teglar bilan **test to‘plamida** qayta baholash; ko‘p urug‘li (seed) tajribalar; global normallash (Kirchhoff matritsa-daraxt teoremasi, $Z(x) = \det \widehat{\mathbf{L}}$); BERTbek va FastText+BERT birikmasi.

684 gap	$+2{,}6$ **LAS**	**54,1 vs 45,4**	**63,8 LAS**
UzUDT korpusi	oxirgi qism so‘z ($p \approx 0{,}002$)	CLE va o‘tish usuli (test LAS)	eng yaxshi (oltin teglar, val.)

Natijalarning aprobatsiyasi va e'lon qilinganligi

1. Matlatipov, S., Aripov, M., Bobokandov, M., & Matlatipov, G. (2026). Annotated Universal Dependencies dataset for literary and educational Uzbek texts. *Data in Brief*, *66*, Article 112857. https://doi.org/10.1016/j.dib.2026.112857 **[Scopus, Q1]**
2. Matlatipov, S. (2027). Towards robust Uzbek neural dependency parsing: Cross-treebank training. In E. Cabrio & E. Monteiro (Eds.), *Natural language processing and information systems* (pp. 204–218). Springer Nature Switzerland. https://doi.org/10.1007/978-3-032-29532-3_15 **[Scopus, Q2]**
3. Matlatipov, S., Ismoilova, M., & Aripov, M. (2026). Graph-based vs transition-based dependency parsing for Uzbek. In *2026 IEEE 27th International Conference of Young Professionals in Electron Devices and Materials (EDM)* (pp. 1–6). IEEE. https://doi.org/10.1109/EDM69524.2026.11632281 **[Scopus, Q3]**
4. Matlatipov, S. G., & Aripov, M. (2026). UzUDT: Uzbek Universal Dependencies treebank. In S. Piperidis, N. Bel, H. van den Heuvel, N. Ide, S. Krek, & A. Toral (Eds.), *Proceedings of the Fifteenth Language Resources and Evaluation Conference* (pp. 11642–11649). ELRA Language Resource Association. https://doi.org/10.63317/2uedjqezjxn5 **[Scopus]**
5. Matlatipov, S., & Kurbanova, M. (2026). A gold-standard dependency treebank for the Uzbek language. In *2026 IEEE Ural-Siberian Conference on Biomedical Engineering, Radioelectronics and Information Technology (USBEREIT)* (pp. 1–3). IEEE. https://doi.org/10.1109/USBEREIT70063.2026.11580531 **[Scopus]**
6. Matlatipov, S. (2026). Neural morphosyntactic tagging and dependency parsing for Uzbek. In *2026 IEEE Ural-Siberian Conference on Biomedical Engineering, Radioelectronics and Information Technology (USBEREIT)* (pp. 1–4). IEEE. https://doi.org/10.1109/USBEREIT70063.2026.11580890 **[Scopus]**

E’tiboringiz uchun rahmat!

Matlatipov Sanatbek

Mirzo Ulug‘bek nomidagi O‘zbekiston Milliy universiteti

s.matlatipov@nuu.uz SanatbekMatlatipov/robust-parsing-uzbek